ADVANCED MACHINE LEARNING · BEFORE YOU BEGIN

Set Up Your Machine-Learning Workshop

About an hour of setup to do **before the first session**, most of it waiting for downloads: Python 3.12 or newer, a virtual environment with the course packages (including PyTorch), a notebook tool, the Pinecrest Outfitters data, and the image data the deep-learning sessions use. These are the same steps as Lesson 0.1 on the course site (rays-python-machine-learning.netlify.app), so if you get stuck, that page shows every step with its expected output. **Took Machine Learning Foundations?** Your environment is ready: skip to Step 6 and run the check, then do Step 7.

⏱️ ~60 min, mostly downloads | Assumes Foundations or equivalent | Windows · Mac · Linux | No GPU needed

> **PLEASE DO THIS AT HOME, NOT IN CLASS**
>
> The packages are several hundred megabytes and the image data another 175 MB. Venue Wi-Fi is rarely up to twenty people doing that at once. If a step fails, keep going with the others and bring the **exact error message** (a screenshot is fine) to Session 1: part of the first half hour is set aside for the environment check.

WHAT YOU NEED

- **A laptop** (Windows, macOS or Linux) and its **charger**, with **permission to install software** and about **3 GB of free disk space** (the environment about 2 GB, the image data about 330 MB). 8 GB of memory is comfortable. Chromebooks, tablets and locked work laptops cannot run the course comfortably: talk to your instructor first.
- **Python 3.12 or newer**, a **terminal** (PowerShell on Windows, Terminal on macOS), and **JupyterLab** (installed with the course packages) or **VS Code**.
- **Two downloads** from the course site: `requirements.txt` and `pinecrest-ml-data.zip` (1.4 MB). Your instructor sends `check_environment.py` with this guide.
- **No graphics card (GPU)**, accounts, API keys or paid software. Every lesson runs on an ordinary laptop processor.
- **A learning journal**: a plain text file called `journal.md` or a paper notebook. Every lesson ends with a journal prompt.

AM I READY FOR THIS COURSE?

Advanced Machine Learning follows **Machine Learning Foundations** (Modules 0 to 6 of the same course site). You are ready if you can, without looking things up:

- ☐ Split a table into training and test rows (stratified, or by time) and say why the test set is used once
- ☐ Build a scikit-learn `Pipeline` with a `ColumnTransformer` for numeric and categorical columns, and say why the scaler belongs inside it
- ☐ Cross-validate on the training rows only and read the mean and the spread of the folds
- ☐ Compare a model with a baseline (`DummyClassifier`) on a metric that fits the decision: ROC AUC, recall, precision, MAE
- ☐ Choose a decision threshold from costs and benefits (an offer costs $15 and a saved churner is worth $60: break-even 0.25)
- ☐ Train and tune a random forest or gradient-boosting model, and read `predict_proba`

Two or more unchecked? Review Modules 1, 2, 5 and 6 of the course site before Session 1. Session 1 opens with a short, ungraded refresher on these same skills, so your instructor knows where to help.

SET UP IN 7 STEPS

1 Python 3.12 or newer

~5 min

Open a terminal and check what you have. A higher number than 3.12 is fine.

SYSTEM	CHECK	MISSING OR OLDER THAN 3.12?
Windows	`py --version`	Install from **python.org/downloads**. In the classic installer, tick **"Add python.exe to PATH"** on the first screen.
macOS	`python3 --version`	The macOS installer from python.org, or with Homebrew: `brew install python@3.12`
Linux (Debian, Ubuntu)	`python3 --version`	`sudo apt install python3 python3-venv python3-pip`. Ubuntu 24.04 and newer give you 3.12. On an older release, install `python3.12` and `python3.12-venv` and use `python3.12` wherever Step 2 says `python3`.

2 A project folder and a virtual environment

~5 min

Everything for the course lives in one folder, `pinecrest-ml`, with its own environment inside it. Run these one line at a time:

	WINDOWS (POWERSHELL)	MACOS / LINUX
Folder	`mkdir pinecrest-ml` then `cd pinecrest-ml`	
Create	`py -m venv .venv`	`python3 -m venv .venv`
Activate	`.venv\Scripts\Activate.ps1`	`source .venv/bin/activate`
Check	Your prompt now starts with `(.venv)`. From here on, plain `python` works on every system.	

> **ACTIVATION IS PER TERMINAL WINDOW**
> Every time you sit down to work: open a terminal, `cd pinecrest-ml`, activate, look for `(.venv)`. Forgetting is the number-one cause of `ModuleNotFoundError`.

3 Install the course packages (PyTorch first on Linux)

~10–20 min

Download **requirements.txt** from the course home page into `pinecrest-ml`. With `(.venv)` showing:

Linux only, first: the default PyTorch download on Linux bundles several gigabytes of GPU libraries you will not use. Install the small CPU build first, and the main install keeps it:

```
python -m pip install torch~=2.14.0 torchvision~=0.29.0 --index-url https://downlo
```

Everyone (on Windows and macOS this is the only line; the default PyTorch there is already the right size):

```
python -m pip install -r requirements.txt
```

It installs about two dozen packages: NumPy, pandas, Matplotlib, SciPy, scikit-learn 1.9, XGBoost, LightGBM, imbalanced-learn, Optuna, umap-learn, SHAP, LIME, PyTorch and torchvision, MLflow, skops, FastAPI, uvicorn, httpx and JupyterLab, plus their dependencies. Success ends with a line starting **Successfully installed**.

4 The course data

~2 min

Download **pinecrest-ml-data.zip** (1.4 MB) from the course home page into `pinecrest-ml`, then unzip it there. This works on every system:

```
python -m zipfile -e pinecrest-ml-data.zip .
```

The `.` means "extract here". You now have a `data/` folder with seven CSV files (customers, weekly demand, payments, reviews, products, purchases, daily store sales), a README and the program that generated them. Pinecrest Outfitters is fictional: every customer, product and number is invented.

```
pinecrest-ml/
├── .venv/              the environment (never edit by hand)
├── data/               customers_churn.csv, transactions.csv, ... and README.md
│   ├── FashionMNIST/   (after Step 7)
│   └── eurosat/        (after Step 7)
├── labs/               (handed out in Session 1)
└── requirements.txt
```

In Session 1 your instructor hands out a `labs/` folder: it goes inside `pinecrest-ml` too, next to `data/`. Every lab finds the data by itself from there.

5 JupyterLab or VS Code

~5 min

- **JupyterLab** (already installed): from `pinecrest-ml` with the environment active, run `jupyter lab`. A browser tab opens; click **Python 3 (ipykernel)** to make a notebook. Stop it with **Ctrl+C** in the terminal.
- **Or VS Code**: install the **Python** and **Jupyter** extensions, **File ▸ Open Folder** on `pinecrest-ml`, and pick the kernel and interpreter that mention `.venv`.

6 Run the environment check

~2 min

Save `check_environment.py` in `pinecrest-ml` (next to `data/`) and run it with the environment active. The first run takes up to a minute: PyTorch and MLflow are slow to import the first time.

```
python check_environment.py
```

It checks five things: Python is 3.12 or newer, an environment is active, every course package imports (with its version, PyTorch and torchvision included), the data folder has all seven files with the right row counts, and whether the image data and pretrained networks are on this computer yet. Its last line is one of three:

- **READY for Advanced Machine Learning**: everything, images included.
- **READY for Sessions 1 to 4**: all set except the image data. Do Step 7.
- **NOT READY**, followed by a numbered list of exactly what to fix.

"PyTorch device: CPU only (expected)" is the normal line. "GPU found" is fine too: the course still runs on the processor so your numbers match the pages.

7 Download the image data and pretrained networks

~5–15 min

Sessions 6 to 9 use two public image collections and two networks already trained on millions of photos. Download them **once, at home, before Session 5**, on a good connection:

```
python check_environment.py --download
```

(Once your instructor has handed out the labs folder, the same file is at `labs/module-01/l7_0-orientation/check_environment.py`.) It downloads what is missing, then checks again; the last line should now say **READY for Advanced Machine Learning**.

WHAT	DOWNLOAD	WHERE IT GOES	FIRST USED
Fashion-MNIST (70,000 clothing pictures, 28 × 28 pixels)	about 30 MB (about 86 MB on disk)	`pinecrest-ml/data/FashionMNIST/`	Session 6 (Lesson 9.3); CNNs in Session 8
EuroSAT (27,000 satellite tiles, 10 land-cover classes)	94 MB (about 186 MB on disk)	`pinecrest-ml/data/eurosat/`	Session 9 (Lesson 10.5)
ResNet-18 pretrained weights	45 MB	PyTorch's cache (below)	Session 9
MobileNetV3-small pretrained weights	10 MB	PyTorch's cache (below)	Session 9

PyTorch's cache is a folder in your home folder: `~/.cache/torch/hub/checkpoints/` on macOS and Linux, `%USERPROFILE%\.cache\torch\hub\checkpoints\` on Windows (the files are `resnet18-f37072fd.pth` and `mobilenet_v3_small-047dcff4.pth`). If the environment variable `TORCH_HOME` is set, the cache is `$TORCH_HOME/hub/checkpoints` instead. Deleting `.venv` does not delete the data or the weights.

> **NO CONNECTION GOOD ENOUGH?**
> Your instructor can bring the four items on a USB stick. Copy `FashionMNIST/` and `eurosat/` into your `pinecrest-ml/data/` folder, and the two `.pth` files into the PyTorch cache folder above (create it if it does not exist). Then run `python check_environment.py` again: it should say READY for Advanced Machine Learning without downloading anything.

✅ YOU'RE READY

Reply to your instructor with the last line of the check. **READY for Advanced Machine Learning** means you are set for the whole course. **READY for Sessions 1 to 4** is fine for Session 1, as long as Step 7 is done before Session 5. **NOT READY**? Try the fixes it lists and the table below, then send the whole output.

COMMON INSTALL PROBLEMS

YOU SEE…	TRY THIS
Linux: pip starts downloading files named `nvidia-…` (gigabytes)	Press Ctrl+C. Run the CPU-only PyTorch line from Step 3 first, then the requirements line again.
Windows: `python` is not recognized, or it opens the Microsoft Store	Before an environment is active, Windows only knows `py`. Create the environment with `py -m venv .venv`; once it is active, `python` works.
PowerShell: "running scripts is disabled on this system"	Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once, answer Y, and activate again.
Debian/Ubuntu: "ensurepip is not available"	`sudo apt install python3-venv python3-pip` (or `python3.12-venv`), delete the half-made `.venv` folder, create it again.
pip: "Could not find a version that satisfies the requirement…"	Check `python --version`: the course needs 3.12 or newer. Install it, delete `.venv`, then repeat Steps 2 and 3.
`ModuleNotFoundError: No module named 'torch'` (or `shap`, `umap`)	The environment is not active in this terminal (no `(.venv)`), or the notebook uses another kernel: start `jupyter lab` from the activated terminal, or choose the `.venv` kernel in VS Code. Still missing: run the requirements line again.
`import torch` warns "Failed to initialize NumPy"	A NumPy was installed outside the requirements file. Run `python -m pip install -r requirements.txt` again in the activated environment.
Windows: `import torch` fails with a DLL error	Install the latest Microsoft Visual C++ Redistributable (x64) from Microsoft's site, then restart the terminal.
The check says "data folder not found"	Python is working in another folder, or the zip was unpacked into a nested folder. You need `pinecrest-ml/data/customers_churn.csv`; run from `pinecrest-ml`.

YOU SEE…	TRY THIS
`--download` stops with a network error, or is very slow	Run it again later on a better connection: what finished is kept. Or ask your instructor for the USB copy (Step 7).
The image data downloads again inside a lesson	The notebook is running in another folder, so `data/` means a different folder. Start JupyterLab from `pinecrest-ml` (check with `print(Path.cwd())`).
The first `import shap` or `import umap` sits silent for 10 to 20 seconds	Normal: they compile parts of themselves the first time. The second import is fast.

Environment badly broken? Delete the `.venv` folder and repeat Steps 2 and 3. Your notebooks, data and downloaded images are not inside it, so nothing of yours is lost.

THE COURSE AT A GLANCE

SESSIONS	YOU'LL LEARN
1–2	Explaining models: permutation importance, partial dependence, SHAP, LIME, a fairness audit
2–5	Clustering: K-Means, choosing k, hierarchical, density-based and mixture models; anomaly detection for fraud
5–6	Dimensionality reduction: PCA, LDA, t-SNE and UMAP (Checkpoint 1 in Session 6)
7–9	Deep learning with PyTorch: a network from scratch, CNNs on Fashion-MNIST, sequence models, transfer learning on EuroSAT
9–10	Text classification and recommendations; reproducibility and experiment tracking with MLflow
11	Serving and monitoring a model with FastAPI; an end-to-end churn case study; you choose your capstone
12–13	Capstone studio and showcase: your own Pinecrest ML project, with a model card

13 sessions of 3 hours. Between sessions you finish each lesson's project, its exercises, quiz and journal entry on the course site: about two to three hours a week.

FREQUENTLY ASKED

DO I NEED A GPU?

No. Every lesson and lab was built and timed on an ordinary laptop processor; the longest training run in class takes about a minute. If you have an NVIDIA GPU you may try the lessons' GPU notes at home, but in class everyone stays on the CPU so results match the pages and your neighbors.

CAN I USE ANACONDA, COLAB OR A SETUP I ALREADY HAVE?

Yes, if you end up with Python 3.12 or newer and the versions in `requirements.txt`, and the check says READY. Demos in class use a `.venv`; a hosted notebook service also needs the data folder and the image data uploaded.

IS THE PINECREST DATA REAL? AND THE IMAGES?

Pinecrest is not real: it was generated for the course (`data/make_ml_data.py` rebuilds it). Fashion-MNIST (Zalando Research) and EuroSAT (built from Copernicus Sentinel-2 satellite images) are real public datasets released under the MIT license; you download them from their sources in Step 7.

WHAT IF A STEP FAILS AND I CANNOT FIX IT?

Come anyway, with the output of `check_environment.py`. Session 1 opens with an environment check and help.

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